

1. If X has Support $X \in S = [1, \infty)$, find the constant c (as a function of θ) that makes $f(x) = c e^{-x/\theta}$ a valid PDF.

$$\begin{aligned} \text{A: } \int_1^{\infty} f(x) dx &= 1 \Rightarrow \int_1^{\infty} c e^{-x/\theta} dx = 1 \Rightarrow c \cdot \left[\frac{e^{-x/\theta}}{-1/\theta} \right]_1^{\infty} = 1 \\ &\Rightarrow c \cdot \left[\frac{e^{-\infty/\theta}}{-1/\theta} - \frac{e^{-1/\theta}}{-1/\theta} \right] = 1 \Rightarrow c \cdot \frac{e^{-1/\theta}}{1/\theta} = 1 \\ &\Rightarrow c \cdot \theta e^{-1/\theta} = 1 \Rightarrow c = \theta^{-1} e^{1/\theta} \end{aligned}$$

2. Assume that $X \sim U(a, b)$ so the Support is $S = [a, b]$ and the PDF is $f(x) = 1/(b-a)$ for any $x \in S$.

(a) prove that this is a valid PDF.

$$\text{A: } f(x) = \frac{1}{b-a}, x \in [a, b] \text{ Here } b > a \text{ (otherwise } X \text{ and } S \text{ are invalid)}$$

$$\text{So, } f(x) > 0 \quad \forall x \in S \quad \text{and} \quad f(x) = 0 \quad \forall x \notin S$$

$$\text{Overall, } f(x) \geq 0.$$

$$\text{Now, } \int_S f(x) dx = \int_a^b \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b 1 dx = \frac{b-a}{b-a} = 1.$$

So, $f(x)$ is a valid PDF.

(b) Derive the mean and variance of X .

$$\begin{aligned} \text{A: } E(X) &= \int_a^b x \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x dx = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b \\ &= \frac{1}{b-a} \cdot \frac{b^2 - a^2}{2} = \frac{a+b}{2} \end{aligned}$$

$$V(X) = E(X^2) - E(X)^2$$

$$\begin{aligned} E(X^2) &= \int_a^b x^2 f(x) dx = \int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{b-a} \int_a^b x^2 dx \\ &= \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3} = \frac{(b-a)(b^2 + ab + a^2)}{3(b-a)} \end{aligned}$$

$$\begin{aligned} V(X) &= \frac{b^2 + ab + a^2}{3} - \left(\frac{a+b}{2} \right)^2 = \frac{b^2 + ab + a^2}{3} - \frac{a^2 + 2ab + b^2}{4} \\ &= \frac{1}{12} [4b^2 + 4ab + 4a^2 - 3a^2 - 6ab - 3b^2] = \frac{1}{12} [a^2 - 2ab + b^2] = \frac{1}{12} (a-b)^2 \end{aligned}$$

3. Expert Knowledge dictates that a parameter must be positive and that its prior distribution should have the mean 5 and Variance 3. Find a prior distribution that satisfies these constraints.

A: This problem does not have a unique solution, rather just an example needs to be provided.

Because, it is positive, one good choice would be a gamma distribution.

Now, $\pi(\theta) \equiv \text{Gamma}(a, b)$

$$E(\theta) = \frac{a}{b}, \quad V(\theta) = \frac{a}{b^2} \quad [\text{check Lecture 1}]$$

$$\text{So, } \frac{a}{b} = 5, \quad \frac{a}{b^2} = 3 \quad \Rightarrow \quad \frac{\frac{a}{b}}{\frac{a}{b^2}} = \frac{5}{3} \Rightarrow b = \frac{5}{3}$$

$$\text{So, } a = 5b = 5 \times \frac{5}{3} = \frac{25}{3}$$

$$\boxed{a = \frac{25}{3}, \quad b = \frac{5}{3}}$$

4. Joint PMF of X_1 and X_2 is

x_1	x_2	$P(X_1=x_1, X_2=x_2)$
0	0	0.15
1	0	0.15
2	0	0.15
0	1	0.15
1	1	0.20
2	1	0.20

(a) Compute the marginal distribution of X_1 .

A: ~~PMF~~ X_1 can take 3 values: 0, 1, 2. X_2 takes 2 values: 0, 1.

$$P(X_1=0) = \sum_{x_2 \in \{0,1\}} P(X_1=0, X_2=x_2) = P(X_1=0, X_2=0) + P(X_1=0, X_2=1) = 0.15 + 0.15 = 0.30$$

$$P(X_1=1) = \sum_{x_2 \in \{0,1\}} P(X_1=1, X_2=x_2) = P(X_1=1, X_2=0) + P(X_1=1, X_2=1) = 0.15 + 0.20 = 0.35$$

$$P(X_1=2) = \sum_{x_2 \in \{0,1\}} P(X_1=2, X_2=x_2) = P(X_1=2, X_2=0) + P(X_1=2, X_2=1) = 0.15 + 0.20 = 0.35$$

(b) Compute the marginal distribution of X_2 .

A: X_2 takes 2 values: 0, 1.

$$\begin{aligned} P(X_2=0) &= \sum_{x_1 \in \{0,1,2\}} P_n(X_1=x_1, X_2=0) \\ &= P_n(X_1=0, X_2=0) + P_n(X_1=1, X_2=0) + P_n(X_1=2, X_2=0) \\ &= 0.15 + 0.15 + 0.15 = 0.45. \end{aligned}$$

$$\begin{aligned} P(X_2=1) &= \sum_{x_1 \in \{0,1,2\}} P_n(X_1=x_1, X_2=1) \\ &= P_n(X_1=0, X_2=1) + P_n(X_1=1, X_2=1) + P_n(X_1=2, X_2=1) \\ &= 0.15 + 0.20 + 0.20 = 0.55. \end{aligned}$$

(c) Compute the conditional distribution of $X_1|X_2$.

A: X_1 takes values $\{0,1,2\}$, X_2 takes values $\{0,1\}$.

So, when we talk about $P_n(X_1=x_1 | X_2=x_2)$, we have to consider 6 cases.

$$P_n(X_1=0 | X_2=0) = \frac{P_n(X_1=0, X_2=0)}{P_n(X_2=0)} = \frac{0.15}{0.45} = \frac{1}{3}.$$

$$P_n(X_1=1 | X_2=0) = \frac{P_n(X_1=1, X_2=0)}{P_n(X_2=0)} = \frac{0.15}{0.45} = \frac{1}{3}.$$

$$P_n(X_1=2 | X_2=0) = \frac{P_n(X_1=2, X_2=0)}{P_n(X_2=0)} = \frac{0.15}{0.45} = \frac{1}{3}.$$

$$P_n(X_1=0 | X_2=1) = \frac{P_n(X_1=0, X_2=1)}{P_n(X_2=1)} = \frac{0.15}{0.55} = \frac{3}{11}.$$

$$P_n(X_1=1 | X_2=1) = \frac{P_n(X_1=1, X_2=1)}{P_n(X_2=1)} = \frac{0.20}{0.55} = \frac{4}{11}.$$

$$P_n(X_1=2 | X_2=1) = \frac{P_n(X_1=2, X_2=1)}{P_n(X_2=1)} = \frac{0.20}{0.55} = \frac{4}{11}.$$

(d) Compute the conditional distribution of $X_2 | X_1$.

$$Pr(X_2=0|X_1=0) = \frac{Pr(X_1=0, X_2=0)}{Pr(X_1=0)} = \frac{0.15}{0.3} = \frac{1}{2}.$$

$$Pr(X_2=1|X_1=0) = \frac{Pr(X_1=0, X_2=1)}{Pr(X_1=0)} = \frac{0.15}{0.3} = \frac{1}{2}$$

$$Pr(X_2=0|X_1=1) = \frac{Pr(X_1=1, X_2=0)}{Pr(X_1=1)} = \frac{0.15}{0.35} = \frac{3}{7}$$

$$Pr(X_2=1|X_1=1) = \frac{Pr(X_1=1, X_2=1)}{Pr(X_1=1)} = \frac{0.2}{0.35} = \frac{4}{7}$$

$$Pr(X_2=0|X_1=2) = \frac{Pr(X_1=2, X_2=0)}{Pr(X_1=2)} = \frac{0.15}{0.35} = \frac{3}{7}$$

$$Pr(X_2=1|X_1=2) = \frac{Pr(X_1=2, X_2=1)}{Pr(X_1=2)} = \frac{0.2}{0.35} = \frac{4}{7}$$

(e) Are X_1 and X_2 independent? Justify your answer.

$$A: Pr(X_1=0, X_2=0) = \cancel{0.15} 0.15$$

$$Pr(X_1=0) = 0.3, Pr(X_2=0) = 0.45, 0.3 \times 0.45 \neq 0.15$$

$$\text{So, } Pr(X_1=0, X_2=0) \neq Pr(X_1=0) \times Pr(X_2=0)$$

So, X_1 and X_2 are not independent.

5/ If (X_1, X_2) is bivariate normal with $E(X_1) = E(X_2) = 0$,
 $Var(X_1) = Var(X_2) = 1$, and $Corr(X_1, X_2) = \rho$.

(a) Derive the marginal distribution of X_1 .

$$f_X(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}(x_1^2 + x_2^2 - 2\rho x_1 x_2)}, x_1, x_2 \in \mathbb{R}$$

So, the marginal distribution of X_1 (density function) is

$$\begin{aligned} f_{X_1}(x_1) &= \int f_X(x_1, x_2) dx_2 \\ &= \frac{1}{2\pi\sqrt{1-\rho^2}} \int e^{-\frac{1}{2(1-\rho^2)}(x_1^2 + x_2^2 - 2\rho x_1 x_2)} dx_2 \end{aligned}$$

$$\begin{aligned}
 f_{X_1}(x_1) &= \frac{1}{2\pi\sqrt{1-\rho^2}} \int e^{-\frac{1}{2(1-\rho^2)}[(x_2 - \rho x_1)^2 + (1-\rho^2)x_1^2]} dx_2 \\
 &= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \cdot e^{-\frac{1}{2(1-\rho^2)}(1-\rho^2)x_1^2} \times \underbrace{\int \frac{1}{\sqrt{2\pi(1-\rho^2)}} (x_2 - \rho x_1)^2 dx_2}_{\text{density of } N(\rho x_1, 1-\rho^2) \text{ distribution}} \times \cancel{\sqrt{1-\rho^2}} \\
 &= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x_1^2}
 \end{aligned}$$

So, Marginal distribution of X_1 is $X_1 \sim N(0, 1)$. Similarly, $X_2 \sim N(0, 1)$.

(b) Derive the Conditional distribution of $X_1|X_2$.

$$\begin{aligned}
 f_{X_1|X_2=x_2}(x_1) &= \frac{f_X(x_1, x_2)}{f_{X_2}(x_2)} \\
 &= \frac{\frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}(x_1^2 + x_2^2 - 2\rho x_1 x_2)}}{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x_2^2}} \\
 &= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}[x_1^2 + x_2^2 - 2\rho x_1 x_2 - (1-\rho^2)x_2^2]} \\
 &= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}[x_1^2 - 2\rho x_1(\rho x_2) + (\rho x_2)^2]} \\
 &= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}(x_1 - \rho x_2)^2},
 \end{aligned}$$

which is the density of a Normal($\rho x_2, 1-\rho^2$) distribution.

So, $X_{01}|X_2 \sim N(\rho x_2, \boxed{1-\rho^2})$. ^{variance, not SD.}

6. Assume (X_1, X_2) have bivariate PDF

$$f(x_1, x_2) = \frac{1}{2\pi} (1 + x_1^2 + x_2^2)^{-3/2}$$

(a) plot the conditional distribution of $X_1|X_2=x_2$ for $x_2 \in \{-3, -2, -1, 0, 1, 2, 3\}$.

A: For plotting, we need a software. But, let's just derive the expression of $f_{X_1|X_2=x_2}(x_1)$.

$$f(x_1, x_2) = \frac{1}{2\pi} (1 + x_1^2 + x_2^2)^{-3/2}$$

$$f_{X_2}(x_2) = \int \frac{1}{2\pi} (1 + x_1^2 + x_2^2)^{-3/2} dx_1$$

$$= \int \frac{1}{2\pi} \cdot \left[(1 + x_2^2) \left(1 + \frac{x_1^2}{1 + x_2^2} \right) \right]^{-3/2} dx_1$$

$$= \frac{1}{2\pi} (1 + x_2^2)^{-3/2} \int_{-\infty}^{+\infty} \left(1 + \frac{x_1^2}{1 + x_2^2} \right)^{-3/2} dx_1$$

the PDF of a Student's t-distribution (with df v) is

$$f(x) = \frac{\Gamma(\frac{v+1}{2})}{\sqrt{v\pi} \Gamma(\frac{v}{2})} \left(1 + \frac{x^2}{v} \right)^{-\frac{v+1}{2}} \quad , \quad \text{Also, } \int_{-\infty}^{+\infty} f(x) dx = 1$$

We next try to match the forms:

$$f_{X_2}(x_2) = \frac{1}{2\pi} (1 + x_2^2)^{-3/2} \int_{-\infty}^{+\infty} \left[1 + \frac{\left(\frac{x_1}{\sqrt{(1+x_2^2)/2}} \right)^2}{2} \right]^{-\frac{2+1}{2}} d\left(\frac{x_1}{\sqrt{(1+x_2^2)/2}} \right)$$

$$\times \sqrt{(1+x_2^2)/2}$$

$$\Gamma(1.5) = 0.5 \Gamma(0.5) = 0.5 \sqrt{\pi}$$

$$= \frac{1}{2\pi} (1 + x_2^2)^{-3/2} \times \sqrt{(1+x_2^2)/2} \times \frac{\sqrt{2\pi} \times \Gamma(\frac{2}{2})}{\Gamma(\frac{2+1}{2})}$$

$$= \frac{1}{2\pi} (1 + x_2^2)^{-3/2 + 1/2} \cdot \frac{1}{\sqrt{2}} \times \frac{\sqrt{2\pi} \times 1}{0.5 \sqrt{\pi}}$$

$$= \frac{1}{2\pi} \times \frac{\sqrt{2\pi}}{0.5 \sqrt{\pi}} (1 + x_2^2)^{-1}$$

$$= \frac{1}{\pi} (1 + x_2^2)^{-1},$$

So, X_2 follows a Standard Cauchy distribution.

$$\text{So, } f_{X_1|X_2=x_2}(x_1) = \frac{\frac{1}{2\pi} (1 + x_1^2 + x_2^2)^{-3/2}}{\frac{1}{\pi} (1 + x_2^2)^{-1}} = \frac{1}{2} \frac{1 + x_2^2}{(1 + x_1^2 + x_2^2)^{3/2}}$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{1 + x_2^2}} \cdot \frac{(1 + x_2^2)^{3/2}}{(1 + x_1^2 + x_2^2)^{3/2}} = \frac{1}{2} \cdot \frac{1}{\sqrt{1 + x_2^2}} \cdot \left(1 + \frac{x_1^2}{1 + x_2^2} \right)^{-3/2}$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{1+x_2^2}} \left(1 + \frac{x_1^2}{\sqrt{(1+x_2^2)/2}} \right)^{-\frac{2+1}{2}}$$

$$\left[\frac{\Gamma(\frac{2+1}{2})}{\sqrt{2\pi} \Gamma(\frac{2}{2})} = \frac{\Gamma(\frac{3}{2})}{\sqrt{2\pi} \times 1} = \frac{0.5 \sqrt{\pi}}{\sqrt{2\pi} \times 1} = \frac{0.5}{\sqrt{2}} = \frac{1}{2\sqrt{2}} \right]$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{(1+x_2^2)/2}} \times \frac{1}{\sqrt{2}} \left(1 + \frac{(x_1 / \sqrt{(1+x_2^2)/2})^2}{2} \right)^{-\frac{2+1}{2}}$$

$$= \frac{1}{2\sqrt{2}} \cdot \frac{1}{\sqrt{(1+x_2^2)/2}} \left(1 + \frac{(x_1 / \sqrt{(1+x_2^2)/2})^2}{2} \right)^{-\frac{2+1}{2}}$$

$$= \frac{\Gamma(\frac{2+1}{2})}{\sqrt{2\pi} \Gamma(\frac{2}{2})} \cdot \frac{1}{\sqrt{(1+x_2^2)/2}} \left(1 + \frac{(x_1 / \sqrt{(1+x_2^2)/2})^2}{2} \right)^{-\frac{2+1}{2}}$$

$$X_1 | X_2 = x_2 \sim \sqrt{\frac{1+x_2^2}{2}} \cdot t_2$$

So, $\pi(X_1 | X_2 = x_2)$ depends on X_2 . So, X_1 and X_2 are not independent.

X_1 and X_2 are identically distributed as standard Cauchy distributions. So, $E(X_1)$ and $E(X_2)$ does not exist.

But, the appearance still possibly makes sense, as is it not possible to judge by just looking at the plot.
the existence of correlation

$E(X_1 | X_2 = x_2) \propto \sqrt{1+x_2^2}$. Because it appears is symmetric around 0 and the correlation would appear zero.

Besides, $f(x_1, x_2) = \frac{1}{2\pi} \left(1 + \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}' \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right)^{-3/2}$

Correlation matrix

Contour plots are circular. As density is a function of $(x^2 + y^2)$.

7/ The 2017 auto theft rate was 135 per 10,000 ~~rate~~ residents in Raleigh. Compared to 214 per 10,000 residents in Durham. Assuming Raleigh's population is ^{twice} as large as Durham and a car has been stolen somewhere in the triangle (one of these areas: Raleigh and Durham), what is the probability it was stolen in Raleigh?

$$A: \Pr(\text{Theft} | \text{Raleigh}) = \frac{135}{10,000}$$

$$\Pr(\text{Theft} | \text{Durham}) = \frac{214}{10,000}$$

$$\Pr(\text{Raleigh}) = \frac{2}{3}, \Pr(\text{Durham}) = \frac{1}{3}$$

$$\begin{aligned} \Pr(\text{Raleigh} | \text{Theft}) &= \frac{\Pr(\text{Theft} | \text{Raleigh}) \Pr(\text{Raleigh})}{\Pr(\text{Theft} | \text{Raleigh}) \Pr(\text{Raleigh}) + \Pr(\text{Theft} | \text{Durham}) \Pr(\text{Durham})} \\ &= \frac{\frac{135}{10,000} \times \frac{2}{3}}{\frac{135}{10,000} \times \frac{2}{3} + \frac{214}{10,000} \times \frac{1}{3}} = \frac{2 \times 135}{2 \times 135 + 214} \\ &= \frac{270}{270 + 214} = \frac{270}{484} = 0.5578. \end{aligned}$$

8/ Your daily commute is distributed uniformly ~~dis~~ between 15 and 20 minutes if there is no convention in the Downtown. However, are scheduled for roughly 1 in 4 days, and your commute time is distributed uniformly from 15 and 30 minutes if there is a convention. let Y = Commute time this morning.

(a) What is the probability that there was a convention in our Downtown given $Y = 18$?

A: So, Support of $Y = [15, 30]$.

$$\Pr(\text{Convention}) = \frac{1}{4}, \Pr(\text{No Convention}) = \frac{3}{4}$$

$$Y | \text{No Convention} \sim U(15, 20), f_0(y) = \frac{1}{5} \mathbb{I}(y \in (15, 20))$$

$$Y | \text{Convention} \sim U(15, 30), f_1(y) = \frac{1}{15} \mathbb{I}(y \in (15, 30))$$

$$f(y) = \Pr(\text{No Convention}) f_0(y) + \Pr(\text{Convention}) f_1(y)$$

$$= \frac{3}{4} \times \frac{1}{5} \mathbb{I}(y \in (15, 20)) + \frac{1}{4} \times \frac{1}{15} \mathbb{I}(y \in (15, 30))$$

$$= \frac{3}{20} \mathbb{I}(y \in (15, 20)) + \frac{1}{60} \mathbb{I}(y \in (15, 30))$$

↑ indicator function

$$\int_{15}^{30} f(y) dy = \int_{15}^{20} f(y) dy + \int_{20}^{30} f(y) dy = \frac{3}{20} \times 5 + \frac{1}{60}$$

$$= \int_{15}^{20} \left(\frac{3}{20} + \frac{1}{60} \right) dy + \int_{20}^{30} \frac{1}{60} dy$$

$$= \frac{10}{60} \times (20-15) + \frac{1}{60} \times (30-20)$$

$$= \frac{10 \times 5}{60} + \frac{1 \times 10}{60} = 1$$

$$f(y) = \frac{3}{20} \mathbb{I}(y \in (15, 20)) + \frac{1}{60} \mathbb{I}(y \in (15, 30))$$

is a valid PDF.

$$\text{So, } \pi(\text{Convention} | Y=18) = \frac{f(18 | \text{Convention}) \times \text{Prob}(\text{Convention})}{f(18 | \text{Convention}) \times \text{Prob}(\text{Conv.}) + f(18 | \text{No Conv.}) \times \text{Pr}(\text{No Conv.})}$$

$$= \frac{\frac{1}{15} \times \frac{1}{4}}{\frac{1}{15} \times \frac{1}{4} + \frac{1}{5} \times \frac{3}{4}}$$

$$= \frac{1/60}{\frac{1}{60} + \frac{3}{20}} = \frac{1}{10} = 0.1$$

What is the prob. that there was a Convention in the Downtown given $Y=28$?

$$\pi(\text{Convention} | Y=28) = \frac{f(28 | \text{Convention}) \times \text{Pr}(\text{Conv.})}{f(28 | \text{Conv.}) \text{Pr}(\text{Conv.}) + f(28 | \text{No Conv.}) \text{Pr}(\text{No Conv.})}$$

$$= \frac{\frac{1}{15} \times \frac{1}{4}}{\frac{1}{15} \times \frac{1}{4} + 0 \times \frac{3}{4}} = 1$$

The answer is natural, as if there ~~was~~ ^{no} Convention, the commute would be less than 20. So, because commute > 20 , we know the ~~no~~ convention is happening certainly.

2. For this problem, pretend that we are dealing with a language with a six-word dictionary.

$\{\text{fun, sun, sit, sat, fan, for}\}$

An extensive study of literature written in this language reveals that all words are equally likely except that "for" is α times as likely as the other words. Further study reveals that:

- Each keystroke is an error with probability θ .
- All letters are equally likely to produce errors.
- Given that a letter is typed incorrectly, it is equally likely to be any other letter.
- Errors are independent across letters.

prob. of correctly typing a word is $(1-\theta)^3$.

prob. of typing "pun" or "fen" when intending to type "fun" is $\theta(1-\theta)^2$.

prob. of typing "foo" or "nnn" when intending to type "fun" is $\theta^2(1-\theta)$.

For each Use Bayes' rule to develop a simple spell checker for this language. For each typed words "Sun", "the", "foo", give the probability that each word in the dictionary was the intended word. Perform this for the parameters:

(a) $\alpha = 2$ and $\theta = 0.1$.

(b) $\alpha = 50$ and $\theta = 0.1$.

(c) $\alpha = 2$ and $\theta = 0.95$.

~~Assume~~ A: All words are of 3 letters. So, #keystroke error $\sim \text{Bin}(3, \theta)$.

word: "Sun" $\rightarrow$

<u>fun</u>	sun	<u>sit</u>	<u>sat</u>	<u>fan</u>	<u>for</u>
1	0	2	2	2	3

"the" $\rightarrow$

<u>fun</u>	<u>sun</u>	<u>sit</u>	<u>sat</u>	<u>fan</u>	<u>for</u>
3	3	3	3	3	3

"foo" $\rightarrow$

<u>fun</u>	<u>sun</u>	<u>sit</u>	<u>sat</u>	<u>fan</u>	<u>for</u>
2	3	3	3	2	1

a) $\alpha = 2 \rightarrow$

	"fun"	"Sun"	"sit"	"sat"	"fan"	"for"	
Weights $\rightarrow$	1	1	1	1	1	2	$\rightarrow \text{total} = 7$
prob $\rightarrow$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{2}{7}$	

We will just do one case:

"Sun" $\rightarrow$

Likelihood $\rightarrow$

$$\binom{3}{1} \theta (1-\theta)^2 \left| \binom{3}{0} \theta^0 (1-\theta)^3 \right| \binom{3}{2} \theta^2 (1-\theta) \left| \binom{3}{2} \theta^2 (1-\theta) \right| \binom{3}{3} \theta^3 (1-\theta)^0$$

$$= 3\theta (1-\theta)^2 = (1-\theta)^3 = 3\theta^2 (1-\theta) = \theta^3$$

$$Pr("fun" | "Sun") = \frac{Pr("Sun" | "fun") Pr("fun")}{Pr("Sun" | "fun") Pr("fun") + Pr("Sun" | "Sun") Pr("Sun") + \dots + Pr("Sun" | "for") Pr("for")}$$

$$= \frac{3\theta (1-\theta)^2 \times \frac{1}{7}}{\left[3\theta (1-\theta)^2 \times \frac{1}{7} + (1-\theta)^3 \times \frac{1}{7} + 3\theta^2 (1-\theta) \times \frac{1}{7} + 3\theta^2 (1-\theta) \times \frac{1}{7} + \theta^3 \times \frac{2}{7} \right]}$$

[For specific θ , this expression can be calculated.]

Similarly, for all words ... until "for" as follows.

$$Pr("for" | "Sun") = \frac{Pr("Sun" | "for") \times Pr("for")}{Pr("Sun" | "fun") Pr("fun") + \dots + Pr("Sun" | "for") Pr("for")}$$

$$= \frac{\theta^3 \times \frac{2}{7}}{3\theta (1-\theta)^2 \times \frac{1}{7} + \dots + \theta^3 \times \frac{2}{7}}$$

Good exercise to do using R.

When α increases $\rightarrow$ prior for "for" increases and so posterior should also increase.

When θ increases $\rightarrow$ errors are more likely and so words with many errors have higher probability.

10/ $X_1 \sim \text{Bernoulli}(\theta) \rightarrow$ indicator θ for a tree species occupying a forest. θ is the prior occupancy probability. A researcher collects a sample of n trees from the forest and X_2 belong to the species of interest. The model is $X_2 | X_1 \sim \text{Binomial}(n, \lambda X_1)$.

Q5 A: First think of the validity of the model:

If $X_1 = 0 \Rightarrow X_2 \sim \text{Binomial}(n, 0) \Rightarrow X_2 = 0$ w.p. 1.
that is reasonable, because if the species is not occupying the forest, there is no chance that the sample would contain any sample of that species.

otherwise, if $X_1 = 1 \Rightarrow X_2 \sim \text{Binomial}(n, \lambda)$

$\Rightarrow X_2 \Rightarrow \#$ Binomial is a very reasonable model on the number of samples out of n samples that belong to that species.

So, perfect model!

$$(a) \Pr(X_1 = X_2 = 0)$$

$$= \Pr(X_1 = 0) \Pr(X_2 = 0 | X_1 = 0)$$

$$= (1 - \theta) \times 1$$

$$= (1 - \theta)$$

$$X_1 = \begin{cases} 1 & \text{w.p. } \theta \\ 0 & \text{w.p. } 1 - \theta \end{cases}$$

$$X_2 | X_1 = 0 \sim \text{Bin}(n, 0) \\ = 0 \text{ w.p. } 1$$

$$\text{So, } \Pr(X_2 = 0 | X_1 = 0) = 1$$

$$(b) \Pr(X_1 = 0) = 1 - \theta$$

$$X_2 | X_1 = 1 \sim \text{Bin}(n, \lambda)$$

$$(c) \Pr(X_2 = 0)$$

$$= \Pr(X_2 = 0 | X_1 = 0) \Pr(X_1 = 0) + \Pr(X_2 = 0 | X_1 = 1) \Pr(X_1 = 1)$$

$$= 1 \times (1 - \theta) + \binom{n}{0} \lambda^0 (1 - \lambda)^{n-0} \times \theta$$

$$= 1 - \theta + \theta (1 - \lambda)^n$$

$$(d) \Pr(X_1 = 0 | X_2 = 0) = \frac{\Pr(X_2 = 0 | X_1 = 0) \Pr(X_1 = 0)}{\Pr(X_2 = 0)}$$

$$= \frac{1 \times (1 - \theta)}{1 - \theta + \theta (1 - \lambda)^n} = \frac{1 - \theta}{1 - \theta + \theta (1 - \lambda)^n}$$

$$(e) \Pr(X_2=0 | X_1=0) = 1$$

$$(f) \Pr(X_1=0 | X_2=1) = \frac{\Pr(X_2=1 | X_1=0) \Pr(X_1=0)}{\Pr(X_2=1)}$$

$$\text{Now, } X_2 | X_1=0 \sim \text{Bin}(n, 0) \Rightarrow \Pr(X_2=1 | X_1=0) = 0.$$

$$\text{So, } \Pr(X_1=0 | X_2=1) = 0.$$

$$(g) \Pr(X_2=0 | X_1=1) \overset{\sim}{=} \binom{n}{0} \lambda^0 (1-\lambda)^{n-0} = (1-\lambda)^n$$

$$(h) \text{ [d]: } \Pr(X_1=0 | X_2=0):$$

n samples were drawn, but no sample of that species was observed: So, one possibility is that ~~there is no~~ ~~tree of that species~~ the species does not occupy the forest. Another possibility is that the species actually occupy the forest but was not observed in the sample.

So, ~~the~~ the bigger θ we have, the bigger is the ~~chance~~ chance of the species occupying the forest. That is, it is natural that the chance of not occupying the forest is smaller.

So, keeping λ and n fixed, the term should decrease with θ . and ~~the~~ smaller prior $\equiv$ smaller posterior.

If n is bigger (keeping θ and λ fixed):

Within a bigger sample, no ~~samples~~ ^{observation} belong to that species. $\Rightarrow$ the chance of that species ~~being~~ occupying the forest is even smaller. $\Rightarrow$ ~~the~~ the chance of $X_1=0$ is more.

$$\text{Here, also } n \uparrow, \frac{1-\theta}{1-\theta+\theta(1-\lambda)^n} \uparrow \text{ this,}$$

So, logical in both ways.

Now, ~~the~~ if λ is bigger $\Rightarrow$ if $X_1=1$, there is less chance of X_2 being zero. ~~So~~ Thus, given $X_2=0$, ~~the~~ (which is a less likely event when λ is large), it is more likely ~~to~~ that the species actually does not occupy. Mathematically, $\lambda \uparrow \Rightarrow \text{denominator} \downarrow$
 $\Rightarrow \Pr(X_1=0 | X_2=0) \uparrow$.

So, logical in both ways!

$$(i) \text{ So, } \frac{1-\theta}{1-\theta+\theta(1-\lambda)^n} \geq 0.95$$

$$\Rightarrow \frac{1-0.5}{0.1-0.5+0.5(1-0.1)^n} \geq 0.95$$

$$\Rightarrow \frac{0.5}{0.95} \geq 0.5(1+0.9^n)$$

$$\Rightarrow \frac{1}{0.95} \geq 1+0.9^n$$

$$\Rightarrow 0.9^n \leq \frac{1}{0.95} - 1 = \frac{0.05}{0.95} = \frac{1}{19}$$

$$\Rightarrow n \log(0.9) \leq \log\left(\frac{1}{19}\right) \Rightarrow \frac{n}{10}$$

$$\Rightarrow n \leq \frac{\log}{\log(0.9)}$$

$$\Rightarrow n \log(0.9) \geq \log\left(\frac{1}{19}\right)$$

$$\Rightarrow n \geq \frac{-\log(19)}{\log(0.9)} \quad \text{Note the sign change as } \log(0.9) < 0.$$

$$\text{So, } n = \left\lceil \frac{-\log(19)}{\log(0.9)} \right\rceil \leftarrow \text{Ceiling operator.}$$

11. In a study that uses Bayes methods to forecast the number of species that will be discovered in future years, some researchers report that the number of marine bivalve species discovered in each year between 2010-2015 was 64, 13, 33, 18, 30, and 20.

$Y_t = \#$ species discovered in year t

$Y_t | \lambda \stackrel{iid}{\sim} \text{Poi}(\lambda)$ and $\lambda \sim \text{Unif}(0, 100)$. Plot the posterior.

A: Here we just calculate the posterior. $\underline{Y} = (64, 13, 33, 18, 30, 20)$

$$p(\lambda | \underline{Y}) \propto p(\underline{Y} | \lambda) \pi(\lambda)$$

$$= \prod_{i=1}^6 e^{-\lambda} \frac{\lambda^{y_i}}{y_i!} \times \frac{1}{100} \mathbb{I}(\lambda \in (0, 100))$$

$$\propto e^{-6\lambda} \frac{\lambda^{\sum y_i}}{y_i!} \mathbb{I}(\lambda \in (0, 100))$$

Note that the kernel of $p(\lambda | \underline{Y})$ is similar to a

Gamma distribution with ~~SP~~ Shape parameter $\sum y_i + 1$ and rate parameter 6 , and truncated between 0 and 100.

So, the final density would be

$$p(\lambda | \underline{y}) = \left[\frac{6^{\sum y_i + 1} \lambda^{\sum y_i} e^{-6\lambda}}{\Gamma(\sum y_i + 1)} \right] / \int_0^{100} \left[\frac{6^{\sum y_i + 1} \lambda^{\sum y_i} e^{-6\lambda}}{\Gamma(\sum y_i + 1)} d\lambda \right]$$

(coeffs cancel out)

$$= \frac{\lambda^{\sum y_i} e^{-6\lambda}}{\int_0^{100} \lambda^{\sum y_i} e^{-6\lambda} d\lambda} \leftarrow \text{plot this one.}$$

Gamma density: $X \sim \text{Gamma}(a, b)$

$$\Rightarrow f_X(x) = \frac{b^a}{\Gamma(a)} \cdot x^{a-1} \cdot e^{-bx}$$

12. $\underline{z} = (X, Y) \sim \text{BDVN}(0, 0, 1, 1, \rho)$
 marginal mean of X $\rightarrow$ ρ $\rightarrow$ marginal variance of Y
 marginal variance of X $\rightarrow$ Correlation.
 $\underline{z}_i = (x_i, y_i)$

Observed data: $(-3.3, -2.6), (0.1, -0.2), (-1.1, -1.5), (2.7, 1.5), (2.0, 1.9), (-0.4, -0.3)$.
 $\underline{z}_4, \underline{z}_5, \underline{z}_6$

$$p(\rho | \underline{z}_i, i=1, \dots, 6) \propto \underbrace{\left(\frac{1}{2\pi\sqrt{1-\rho^2}} \right)^6 e^{-\frac{1}{2(1-\rho^2)} \sum_{i=1}^6 (x_i^2 + y_i^2 - 2\rho x_i y_i)}}_{\text{Likelihood}} \times \underbrace{\frac{1}{2} \mathbb{I}(\rho \in (-1, 1))}_{\text{prior}}$$

this is not in any known distributional form.

$$\text{So, } p(\rho | \underline{z}_i) = \frac{(1-\rho^2)^{-3} e^{-\frac{1}{2(1-\rho^2)} \sum_{i=1}^6 (x_i^2 + y_i^2 - 2\rho x_i y_i)}}{\int_{-1}^{+1} (1-\rho^2)^{-3} e^{-\frac{1}{2(1-\rho^2)} \sum_{i=1}^6 (x_i^2 + y_i^2 - 2\rho x_i y_i)} d\rho}$$

13. NDVI (normalized difference vegetation index) is a classifier of land cover based on Satellite data.
 Claim: NDVI $\sim \text{Beta}(25, 10)$ for pixels in a rain forest, NDVI $\sim \text{Beta}(10, 15)$ for pixels that are deforested. Assumption: 10% of the rain forest is deforested.

(a) Plot the PDF of NDVI for forested and deforested pixels.

A: Let $X = \text{NDVI}$, $X \sim \text{Beta}(25, 10)$ for forested pixels.

$$\text{So, } f_{X|\text{forested}}(x) = \frac{\Gamma(35)}{\Gamma(25)\Gamma(10)} x^{25-1} (1-x)^{10-1} \text{ for forested pixels.}$$

$X \sim \text{Beta}(10, 15)$

$$f_{X|\text{deforested}}(x) = \frac{\Gamma(25)}{\Gamma(10)\Gamma(15)} x^{10-1} (1-x)^{15-1} \text{ for deforested pixel}$$

$$\text{overall, } \text{Prob}(\text{forested pixel}) = 0.9$$

$$\text{Prob}(\text{deforested pixel}) = 0.1$$

So, overall,

$$f_X(x) = f_{X|\text{forested}}(x) \text{Prob}(\text{forested}) +$$

$$f_{X|\text{deforested}}(x) \text{Prob}(\text{deforested})$$

$$= 0.9 \frac{\Gamma(35)}{\Gamma(25)\Gamma(10)} x^{25-1} (1-x)^{10-1} + 0.1 \frac{\Gamma(25)}{\Gamma(10)\Gamma(15)} x^{10-1} (1-x)^{15-1}$$

(b) Give an expression of the probability that a pixel is deforested given its NDVI.

$$\begin{aligned} \text{Prob}(\text{deforested} | X=x) &= \frac{f_{X|\text{deforested}}(x) \times \text{Pr}(\text{deforested})}{f_{X|\text{deforested}}(x) \times \text{Pr}(\text{deforested}) + f_{X|\text{forested}}(x) \times (1 - \text{Pr}(\text{deforested}))} \\ &= \frac{0.1 \times \frac{\Gamma(25)}{\Gamma(10)\Gamma(15)} x^{10-1} (1-x)^{15-1}}{0.1 \times \frac{\Gamma(25)}{\Gamma(10)\Gamma(15)} x^{10-1} (1-x)^{15-1} + 0.9 \times \frac{\Gamma(35)}{\Gamma(25)\Gamma(10)} x^{25-1} (1-x)^{10-1}} \end{aligned}$$

(c) Suppose that value is c and range would be $(0, c)$

(Small $X \Rightarrow$ deforestation)

~~Solve~~ $\text{Prob}(\text{deforested} | X=x)$ is a function of x .

and a decreasing function of x .

So, Solve $\text{Prob}(\text{deforested} | X=c) \geq 0.9$

$$\Rightarrow \frac{0.1 \text{ dbeta}(c, 10, 15)}{0.1 \text{ dbeta}(c, 10, 15) + 0.9 \text{ dbeta}(c, 25, 10)} \geq 0.9$$

$$\Rightarrow 0.1 \text{ dbeta}(c, 10, 15) \geq 0.09 \text{ dbeta}(c, 10, 15) + 0.81 \text{ dbeta}(c, 25, 10)$$

$$\Rightarrow 0.01 \text{ dbeta}(c, 10, 15) \geq 0.81 \text{ dbeta}(c, 25, 10)$$

Now, need to solve numerically.

14. $n =$ unknown # Customers.

$Y =$ # Customers that make a purchase

$Y|n \sim \text{Bin}(n, \theta)$, $\theta =$ prob. of purchasing given that the Customer visited the store

Prior: $n \sim \text{Poi}(5)$.

Assume θ is known and n is unknown, plot the posterior distribution of n .

$$\begin{aligned} P(n = n^* | Y=y) &= \frac{P(Y=y | n=n^*) P(n=n^*)}{\sum_{n'=0}^{\infty} P(Y=y | n=n') P(n=n')} \\ &= \frac{\binom{n^*}{y} \theta^y (1-\theta)^{n^*-y} \times e^{-5} \cdot 5^{n^*} / n^*!}{\sum_{n'=0}^{\infty} \binom{n'}{y} \theta^y (1-\theta)^{n'-y} \times e^{-5} \cdot \frac{5^{n'}}{n'!}} \end{aligned}$$

Let's think the effect of θ logically, numerically not possible.

Suppose, $Y=y$ is fixed (given).

If $\theta \uparrow$, prob. of purchasing increases. Now, # customers making purchase remain the same $\Rightarrow n$ decreases stochastically.

So, $\theta \uparrow \Rightarrow P(n=n^* | Y=y)$ weights shifted to smaller values of n^* .

15. Last spring your lab planted 10 seedlings and two survived the winter. Let θ be the prob. that a seedling survives the winter.

(a) Assuming a uniform prior distribution for θ , compute its posterior mean and sd.

A: $Y =$ # seedlings survived the winter. $Y \sim \text{Bin}(10, \theta)$

We observed $Y=2$. Here, prior $\pi(\theta) = \text{Uniform}(0,1) \equiv \text{Beta}(1,1)$

$$P(\theta|Y) \propto f(Y|\theta) \pi(\theta) \propto \theta^2 (1-\theta)^8 \times \theta^{1-1} (1-\theta)^{1-1}$$

$$= \theta^{2+1-1} (1-\theta)^{8+1-1} \propto \text{density of Beta}(3,9)$$

$$So, \theta | Y=2 \sim \text{Beta}(3, 9)$$

$$So, \text{posterior mean } E(\theta | Y=2) = \frac{3}{3+9} = \frac{3}{12} = 0.25.$$

$$\text{posterior sd } SD(\theta | Y=2) = \sqrt{\frac{3 \times 9}{(3+9)^2(3+9+1)}} = \sqrt{\frac{27}{12^2 \times 13}}$$

(b) Assuming the same prior for θ , Compute and Compare the equal-tailed and highest density 95% posterior credible intervals (CI)

So, equal tailed 95% posterior CI is (in terms of R code)

$$(qbeta(0.025, 3, 9), qbeta(0.975, 3, 9))$$

2.5th percentile of Beta(3,9) distribution 97.5th percentile of Beta(3,9) distribution

Next. highest density 95% posterior CI.

Let, it starts at $\rightarrow qbeta(c, 3, 9)$. Then it should end at $qbeta(c+0.95, 3, 9)$

$$So, \text{length} = qbeta(c+0.95, 3, 9) - qbeta(c, 3, 9)$$

We have choose c so that length is minimum.

$$\text{Suppose } c^* = \arg \min_c [qbeta(c+0.95, 3, 9) - qbeta(c, 3, 9)].$$

In R, we might run this minimization over a grid

$c(0.0001, 0.0002, \dots, 0.0499)$. or use "optimize"

function ~~but~~ in R (for project) Final interval: $(qbeta(c^*, 3, 9), qbeta(c^*+0.95, 3, 9))$

(c) If you plant another 10 seedlings next year, what is the posterior predictive probability that at least one will survive the winter?

$$Y^* | \theta \sim \text{Bin}(10, \theta)$$

$$f(Y^* | Y) = \int f(Y^* | \theta) p(\theta | Y) d\theta$$

$$= \int \binom{10}{Y^*} \theta^{Y^*} (1-\theta)^{10-Y^*} \times \frac{\Gamma(12)}{\Gamma(3)\Gamma(9)} \theta^{3-1} (1-\theta)^{9-1} d\theta$$

$$= \binom{10}{Y^*} \times \frac{\Gamma(12)}{\Gamma(3)\Gamma(9)} \int \theta^{Y^*+3-1} (1-\theta)^{10-Y^*+9-1} d\theta$$

$$= \binom{10}{Y^*} \frac{\Gamma(12)}{\Gamma(3)\Gamma(9)} \times \frac{\Gamma(Y^*+3) \Gamma(19-Y^*)}{\Gamma(22)}, \quad Y^* = 0, 1, \dots, 10.$$

$$P_2(Y^* > 0 | Y) = 1 - P_2(Y^* = 0 | Y) = 1 - \binom{10}{0} \frac{\Gamma(12)}{\Gamma(3)\Gamma(9)} \times \frac{\Gamma(3) \Gamma(19)}{\Gamma(22)} = 1 - \frac{\Gamma(12) \Gamma(19)}{\Gamma(9) \Gamma(22)}$$

16/ X_1 and X_2 are binary indicators of failure for two parts of a machine. Independent tests have shown that $X_1 \sim \text{Bernoulli}(1/2)$ and $X_2 \sim \text{Bernoulli}(1/3)$.

Y_1 and Y_2 are binary ~~system~~ indicators of two system failures.

So, using AND/OR gates, $Y_1 = \begin{cases} 1 & \text{if both } X_1=1, X_2=1 \\ 0 & \text{otherwise} \end{cases}$

$Y_2 = \begin{cases} 0 & \text{if both } X_1=0, X_2=0 \\ 1 & \text{otherwise} \end{cases}$

Compute the following probabilities:

(a) The probability that $X_1=1$ and $X_2=1$ given $Y_1=1$.

$Y_1=1$ if $X_1=1$ and $X_2=1$. So, given $Y_1=1$ and $Y_1 \neq 1$ otherwise.

$$P(X_1=1, X_2=1 | Y_1=1) = 1.$$

(b) The probability that $X_1=1$ and $X_2=1$ given $Y_2=1$.

$$Y_2=1 \Rightarrow \begin{cases} X_1=1, X_2=1 \\ X_1=1, X_2=0 \\ X_1=0, X_2=1 \end{cases} \quad \text{One of these cases.}$$

$$\begin{aligned} \text{So, } P(Y_2=0) &= P(X_1=0, X_2=0) \\ &= P(X_1=0) P(X_2=0) \\ &= \frac{1}{2} \times \frac{2}{3} = \frac{1}{3} \end{aligned}$$

No reason to assume X_1 and X_2 are dependent

$$\text{So, } P(Y_2=1) = \frac{2}{3}.$$

$$\text{Now, } P(X_1=1, X_2=1 | Y_2=1) = \frac{P(X_1=1, X_2=1, Y_2=1)}{P(Y_2=1)}$$

$$\begin{aligned} &= \frac{P(X_1=1, X_2=1, Y_2=1)}{P(Y_2=1)} \\ &= \frac{P(X_1=1, X_2=1)}{P(Y_2=1)} = \frac{P(X_1=1) P(X_2=1)}{2/3} = \frac{\frac{1}{2} \times \frac{1}{3}}{2/3} \end{aligned}$$

$$= \frac{1}{2} \times \frac{1}{3} \times \frac{3}{2} = \frac{1}{4}$$

$$\{X_1=1, Y_1=1\} = \{X_1=1, Y_1=1\}$$

(c) Probability that $X_1=1$ given $Y_1=1$.

$$P(X_1=1 | Y_1=1) = \frac{P(X_1=1, Y_1=1)}{P(Y_1=1)} = \frac{P(X_1=1, X_2=1)}{P(Y_1=1)}$$

$$= \frac{\frac{1}{2} \times \frac{1}{3}}{\frac{1}{3}} = \frac{P(X_1=1, X_2=1)}{P(X_1=1, X_2=1)} = 1. \quad \text{Also, logical: } Y_1=1 \Rightarrow X_1=1, X_2=1$$

(a) The probability that $X_1 = 1$ ~~and~~ given $Y_2 = 1$.

$$Pr(X_1 = 1 | Y_2 = 1) = \frac{Pr(X_1 = 1, Y_2 = 1)}{Pr(Y_2 = 1)}$$

$$\begin{aligned} Pr(X_1 = 1, Y_2 = 1) &= Pr(X_1 = 1, X_2 = 0) + Pr(X_1 = 1, X_2 = 1) \\ &= Pr(X_1 = 1) Pr(X_2 = 0) + Pr(X_1 = 1) Pr(X_2 = 1) \\ &= Pr(X_1 = 1) \times [Pr(X_2 = 0) + Pr(X_2 = 1)] \\ &= Pr(X_1 = 1) \\ &= \frac{1}{2} \end{aligned}$$

So, $Pr(Y_2 = 1) = \frac{2}{3}$

So, $Pr(X_1 = 1 | Y_2 = 1) = \frac{1/2}{2/3} = \frac{3}{4}$

17.

Player	Overall prop. (θ_i)	Clutch makes (Y_i)	Clutch attempts (N_i)
P1	0.845	64	75
P2	0.847	72	95
P3	0.880	55	63
P4	0.674	27	39
P5	0.909	75	83
P6	0.898	24	26
P7	0.770	28	41
P8	0.801	66	82
P9	0.802	40	54
P10	0.875	13	16

Clutch: a free throws taken in pressure situations.

The overall proportion is computed using a large sample size, assume it is fixed and analyze the clutch data separately for each player using Bayesian methods. Assume uniform priors.

(a) Describe your model for studying the Clutch Success probability including the likelihood and prior.

~~Q~~ A:- Likelihood: $Y_i | N_i, \theta_i \sim \text{Bin}(N_i, \theta_i)$

Prior: $\theta_i \sim \text{Uniform}(0, 1) \equiv \text{Beta}(1, 1)$.

(b) Plot the posteriors of the Clutch Success probabilities.

~~Plot using~~ Plot using Some Software.

~~Plot using~~ $\theta_i | Y_i \sim \text{Beta}(Y_i + 1, N_i - Y_i + 1)$.

(c) Summarize the posteriors in a table.

~~Report~~ Report Posterior mean, posterior SD, 95% equal-tailed ~~credible~~ posterior credible region.

$$\theta_i | Y_i \sim \text{Beta}(Y_i + 1, N_i - Y_i + 1)$$

$$\text{So, } E(\theta_i | Y_i) = \frac{Y_i + 1}{N_i + 2}, \quad V(\theta_i | Y_i) = \frac{(Y_i + 1)(N_i - Y_i + 1)}{(N_i + 2)^2 (N_i + 3)}$$

$$CI = \left(q_{\text{beta}}(0.025, Y_i + 1, N_i - Y_i + 1), q_{\text{beta}}(0.975, Y_i + 1, N_i - Y_i + 1) \right).$$

(d) Do you find evidence that any of the players have a different Clutch percentage & than overall percentage?

A: Check if the overall proportion falls between CI in part (c).
If not, report that.

(e) Are the results sensitive to prior?

~~Try~~ Try two weakly-informative priors: $\text{Beta}(0.5, 0.5)$, $\text{Beta}(2, 2)$ only for players with small information (less ~~number~~ number of tries, less number of successes), it would affect more.
This is natural.

$$\theta_i | Y_i \sim \text{Beta}(Y_i + 0.5, N_i - Y_i + 0.5) \text{ when } \theta_i \sim \text{Beta}(0.5, 0.5)$$

$$\theta_i | Y_i \sim \text{Beta}(Y_i + 2, N_i - Y_i + 2), \text{ when } \theta_i \sim \text{Beta}(2, 2)$$

18. In the early twentieth century, it was generally agreed that Hamilton and Madison wrote 51 and 14 Federalist Papers, resp. there was dispute over how to attribute 12 other papers between these two authors. In the 51 papers attributed to Hamilton, the word "upon" was used 3.24 times per 1,000 words, compared to 0.23 times per 1,000 words in the 14 papers attributed to Madison.

(a) If the word "upon" is used three times in a disputed text of length 1000 words and we assume the prior prob. 0.5, what is the posterior prob. that the paper was written by Hamilton?

$$A:- \Pr(\text{Hamilton}) = \frac{1}{2}, \quad \Pr(\text{Madison}) = \frac{1}{2}$$

$Y = \# \text{ "upon" used in the text of length 1000 words}$

$$Y | \text{Hamilton} \sim \text{Bin}(1000, \frac{3.24}{1000})$$

$$Y | \text{Madison} \sim \text{Bin}(1000, \frac{0.23}{1000})$$

$$\Pr(Y=3 | \text{Hamilton}) = \binom{1000}{3} \left(\frac{3.24}{1000}\right)^3 \left(1 - \frac{3.24}{1000}\right)^{997}$$

$$\Pr(Y=3 | \text{Madison}) = \binom{1000}{3} \left(\frac{0.23}{1000}\right)^3 \left(1 - \frac{0.23}{1000}\right)^{997}$$

$$\begin{aligned} \Pr(\text{Hamilton} | Y=3) &= \frac{\Pr(Y=3 | \text{Hamilton}) \Pr(\text{Hamilton})}{\Pr(Y=3 | \text{Hamilton}) \Pr(\text{Hamilton}) + \Pr(Y=3 | \text{Madison}) \Pr(\text{Madison})} \\ &= \frac{\cancel{\binom{1000}{3}} \left(\frac{3.24}{1000}\right)^3 \left(1 - \frac{3.24}{1000}\right)^{997} \times \frac{1}{2}}{\cancel{\binom{1000}{3}} \left(\frac{3.24}{1000}\right)^3 \left(1 - \frac{3.24}{1000}\right)^{997} \times \frac{1}{2} + \cancel{\binom{1000}{3}} \left(\frac{0.23}{1000}\right)^3 \left(1 - \frac{0.23}{1000}\right)^{997} \times \frac{1}{2}} \\ &= \frac{3.24^3 (1000 - 3.24)^{997}}{3.24^3 (1000 - 3.24)^{997} + 0.23^3 (1000 - 0.23)^{997}} \end{aligned}$$

(b) Give one assumption you are making in (a) that is likely unreasonable. Justify your answer.

A:- We assume $Y | \text{Hamilton} \sim \text{Bin}(1000, p)$

$\Rightarrow$ Writing Binomial as a Sum of Bernoulli's,

$$Y | \text{Hamilton} = \sum_{i=1}^{1000} X_i, \quad X_i \stackrel{iid}{\sim} \text{Bernoulli}(p)$$

Now, $X_1 = \begin{cases} 1 & \text{if 1st word is "upon"} \\ 0 & \text{if 1st word is not "upon"} \end{cases}$

$X_2 = \begin{cases} 1 & \text{if 2nd word is "upon"} \\ 0 & \text{if 2nd word is not "upon"} \end{cases}$

Now, given that the first word is "upon", it is unlikely that the second word would also be "upon". $\Rightarrow X_i$ are unlikely to be indep.
 $\Rightarrow \sum_{i=1}^{1000} X_i \not\sim \text{Binomial}(1000, p)$.

(c) In (a), if the # instances of "upon" to one, do you expect the posterior prob. to increase, decrease or remain the same? why?

$$Pr(\text{Hamilton} | Y=1) = \frac{3.24 \times (1000 - 3.24)}{3.24 \times (1000 - 3.24) + 0.23 \times (1000 - 0.23)}$$

$$Pr(Y=3 | \text{Hamilton}) = \binom{1000}{3} \left(\frac{3.24}{1000} \right)^3 \left(1 - \frac{3.24}{1000} \right)^{997}$$

$$Pr(Y=1 | \text{Hamilton}) = \binom{1000}{1} \left(\frac{3.24}{1000} \right) \times \left(1 - \frac{3.24}{1000} \right)$$

It is natural that $Pr(Y=1 | \text{Hamilton}) < Pr(Y=3 | \text{Hamilton})$
 $\hookrightarrow 1$ is far from $E(Y) = 3.24$ $\hookrightarrow 3$ is close to $E(Y) = 3.24$

Besides, $Pr(Y=1 | \text{Madison}) = \binom{1000}{1} \left(\frac{0.23}{1000} \right) \left(1 - \frac{0.23}{1000} \right)$

$$Pr(Y=3 | \text{Madison}) = \binom{1000}{3} \left(\frac{0.23}{1000} \right)^3 \left(1 - \frac{0.23}{1000} \right)^{997}$$

It is natural that $Pr(Y=1 | \text{Madison}) > Pr(Y=3 | \text{Madison})$
 $\hookrightarrow 1$ is close to $E(Y) = 0.23$ $\hookrightarrow 3$ is far from $E(Y) = 0.23$

$$Pr(\text{Hamilton} | Y=1) = \frac{Pr(Y=1 | \text{Hamilton})}{Pr(Y=1 | \text{Hamilton}) + Pr(Y=1 | \text{Madison})}$$

$$= \frac{1}{1 + \frac{Pr(Y=1 | \text{Madison})}{Pr(Y=1 | \text{Hamilton})}}$$

We have, $Pr(Y=1 | \text{Hamilton}) < Pr(Y=3 | \text{Hamilton})$
 $Pr(Y=1 | \text{Madison}) > Pr(Y=3 | \text{Madison})$

$$\Rightarrow \frac{Pr(Y=1 | \text{Madison})}{Pr(Y=1 | \text{Hamilton})} > \frac{Pr(Y=3 | \text{Madison})}{Pr(Y=3 | \text{Hamilton})}$$

$$\Rightarrow Pr(\text{Hamilton} | Y=1) < Pr(\text{Hamilton} | Y=3)$$

$\Rightarrow$ the posterior prob. decrease.

(d) If $p(a)$, if we changed the text length to 10,000 words and number of instances of "upon" to 30, do you expect the posterior to increase, decrease or stay the same? Why?

$$A: \Pr(\text{Hamilton} | Y=30) = \frac{3 \cdot 24^{30} (1000 - 3 \cdot 24)^{970}}{3 \cdot 24^{30} (1000 - 3 \cdot 24)^{970} + 0 \cdot 23^{30} (1000 - 0 \cdot 23)^{970}}$$

$$= \frac{1}{1 + \left(\frac{0 \cdot 23}{3 \cdot 24}\right)^{30} \left(\frac{1000 - 0 \cdot 23}{1000 - 3 \cdot 24}\right)^{970}}$$

$$\Pr(\text{Hamilton} | Y=3) = \frac{1}{1 + \left(\frac{0 \cdot 23}{3 \cdot 24}\right)^3 \left(\frac{1000 - 0 \cdot 23}{1000 - 3 \cdot 24}\right)^{997}}$$

Mathematically, we can always compare.

Logically, $Y=30$ is highly unlikely for both Hamilton and Madison. However, it is way more unlikely for Madison.

$$\text{So, } \frac{\Pr(Y=30 | \text{Madison})}{\Pr(Y=30 | \text{Hamilton})} < \frac{\Pr(Y=3 | \text{Madison})}{\Pr(Y=3 | \text{Hamilton})}$$

$$\Rightarrow \Pr(\text{Hamilton} | Y=30) = \frac{1}{1 + \frac{\Pr(Y=30 | \text{Madison})}{\Pr(Y=30 | \text{Hamilton})}}$$

$$> \frac{1}{1 + \frac{\Pr(Y=3 | \text{Madison})}{\Pr(Y=3 | \text{Hamilton})}} = \Pr(\text{Hamilton} | Y=3)$$

(e) Let Y be the number of the observed # instances of "upon" in 1000 words. Compute the posterior prob. the paper was written by Hamilton for each $Y \in \{0, 1, \dots, 20\}$

$$\Pr(\text{Hamilton} | Y=y) = \frac{3 \cdot 24^y (1000 - 3 \cdot 24)^{1000-y}}{3 \cdot 24^y (1000 - 3 \cdot 24)^{1000-y} + 0 \cdot 23^y (1000 - 0 \cdot 23)^{1000-y}}$$

$$\Pr(\text{Madison} | Y=y) = \frac{0 \cdot 23^y (1000 - 0 \cdot 23)^{1000-y}}{3 \cdot 24^y (1000 - 3 \cdot 24)^{1000-y} + 0 \cdot 23^y (1000 - 0 \cdot 23)^{1000-y}}$$

a: plot $\Pr(\text{Hamilton} | Y=y) \Rightarrow$ use some software, R, for example.

Q: Give a rule for the number of instances of 'upon' needed before the paper should be attributed to Hamilton.

$$A:- \Pr(\text{Hamilton} | Y=y) > \Pr(\text{Madison} | Y=y)$$

$$\Rightarrow \frac{\Pr(\text{Hamilton} | Y=y)}{\Pr(\text{Madison} | Y=y)} > 1$$

$$\Rightarrow \frac{3.24^y (1000 - 3.24)^{1000-y}}{0.23^y (1000 - 0.23)^{1000-y}} > 1$$

$$\Rightarrow \left(\frac{3.24}{0.23} \right)^y \left(\frac{1000 - 3.24}{1000 - 0.23} \right)^{1000-y} > 1$$

$$\Rightarrow \left[\frac{3.24 \times (1000 - 0.23)}{0.23 \times (1000 - 3.24)} \right]^y > \left(\frac{1000 - 0.23}{1000 - 3.24} \right)^{1000}$$

$$\Rightarrow y > \frac{1000 \log \left(\frac{1000 - 0.23}{1000 - 3.24} \right)}{\log \left[\frac{3.24 \times (1000 - 0.23)}{0.23 \times (1000 - 3.24)} \right]} = c \text{ (say)}$$

So, if $y > c \Rightarrow$ paper should be attributed to Hamilton.